THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of prime numbers between 20 and 50 is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 9

2. The prime factorisation of 120 has how many prime factors (counting repeats)? 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 5

3. In the Sieve of Eratosthenes up to 50, we need to cross out multiples of primes up to: 1

(a) 7 **(b)** 11 **(c)** 5 **(d)** 25

4. The smallest prime number that is a factor of 143 is: 1

(a) 13 **(b)** 143 **(c)** 1 **(d)** 11

5. Which of the following is divisible by 6? 1

(a) 4220 **(b)** 4219 **(c)** 4218 **(d)** 4221

6. Which pair is a pair of co-prime numbers? 1

(a) 15 and 30 **(b)** 12 and 20 **(c)** 9 and 28 **(d)** 9 and 21

7. Which of these is a pair of twin primes? 1

(a) 43 and 47 **(b)** 29 and 31 **(c)** 31 and 37 **(d)** 37 and 41

8. The number of prime numbers between 40 and 70 is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 9

9. The prime factorisation of 144 has how many prime factors (counting repeats)? 1

(a) 5 **(b)** 7 **(c)** 2 **(d)** 6

10. In the Sieve of Eratosthenes up to 100, we need to cross out multiples of primes up to: 1

(a) 50 **(b)** 5 **(c)** 7 **(d)** 11

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write two prime numbers whose sum is 36. 2

12. Are 18 and 27 co-prime? Give the reason. 2

13. Test whether 5348 is divisible by 4 and by 8. 2

14. Write two prime numbers whose sum is 48. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Use the divisibility test for 11 to check whether 3849 is divisible by 11. 3

16. Draw a factor tree (describe) for 126 and write its prime factorisation. 3

17. Find the first 4 common multiples of 5 and 10. 3

OR

Use the divisibility test for 11 to check whether 6094 is divisible by 11. 3

18. Draw a factor tree (describe) for 150 and write its prime factorisation. 3

OR

Find the first 3 common multiples of 9 and 12. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find all prime numbers between 60 and 100 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. 5

OR

A number is divisible by both 5 and 6. Show that it is divisible by 30 when 5 and 6 are co-prime. Give an example with 90. 5

20. Find all prime numbers between 1 and 30 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. 5

OR

A number is divisible by both 3 and 8. Show that it is divisible by 24 when 3 and 8 are co-prime. Give an example with 96. 5

END OF QUESTION PAPER — Total: 40 Marks

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